

FINAL

Wednesday, Jun. 12, 2019
9:00am–12:00pm

NAME: _____

Student ID: _____

- Don't forget to put your name and student ID.
- Record all your solutions in this answer booklet. Only this answer booklet will be considered in the grading of your exam.
- Be sure to **show all relevant work and reasoning**. A correct answer does not guarantee full credit, and a wrong answer does not guarantee loss of credit. You should clearly but concisely indicate your reasoning.
- In this exam, every “log” we consider is $\log_e$, i.e., $\ln$.

Problem	Your score	Max score
1		10
2		15
3		15
Total		40

Problem 1

Consider two discrete memoryless channels (DMC):

$$P_{Y|X}^{(1)}(\cdot|\cdot) : \mathcal{X}_1 \rightarrow \mathcal{Y}_1$$

$$P_{Y|X}^{(2)}(\cdot|\cdot) : \mathcal{X}_2 \rightarrow \mathcal{Y}_2$$

Let C_1 and C_2 be the capacity of the two channels, respectively. Let $P_{X_1}^*$ and $P_{X_2}^*$ be the optimal input distribution (a distribution achieving the capacity) for the two channels, respectively.

Suppose that the input and output alphabet of the two channels are disjoint, i.e.,

$$\mathcal{X}_1 \cap \mathcal{X}_2 = \emptyset \text{ AND } \mathcal{Y}_1 \cap \mathcal{Y}_2 = \emptyset.$$

Now define a new DMC, which is called the “sum” of these two channels.

$$P_{Y|X}(\cdot|\cdot) : (\mathcal{X}_1 \cup \mathcal{X}_2) \rightarrow (\mathcal{Y}_1 \cup \mathcal{Y}_2)$$

Let

$$P_{Y|X}(y|x) \triangleq \begin{cases} P_{Y|X}^{(1)}(y|x) & \text{if } x \in \mathcal{X}_1, y \in \mathcal{Y}_1 \\ P_{Y|X}^{(2)}(y|x) & \text{if } x \in \mathcal{X}_2, y \in \mathcal{Y}_2 \\ 0 & \text{else.} \end{cases}$$

- a) (5 points) Determine the capacity C of the resulting channel $P_{Y|X}(\cdot|\cdot)$. Write C in terms of C_1 and C_2 . (Hint: Define a Bernoulli random variable

$$Z = \begin{cases} 0, & \text{if } X \in \mathcal{X}_1, \\ 1, & \text{if } X \in \mathcal{X}_2. \end{cases}$$

Notice that the mutual information between X and Y is equal to $I(X; Y) = I(X, Z; Y)$ since Z is determined by X . You may start from here and try to find $p = \Pr(Z = 0)$ that maximizes $I(X, Z; Y)$.)

- b) (5 points) What is the optimal input distribution P_X^* over $\mathcal{X}_1 \cup \mathcal{X}_2$ that achieves the capacity you found in a)? Write it in terms of $P_{X_1}^*$, $P_{X_2}^*$, C_1 and C_2 .

Problem 2

In this problem, we prove Shannon's coding theorem by using a new decoder: maximum mutual information (MMI) decoder.

For a given pair of length- n sequences (x^n, y^n) , we write the joint empirical distribution as $\hat{P}_{(x^n, y^n)}$ and call $I(\hat{P}_{(x^n, y^n)})$ the empirical mutual information of this pair of sequences defined as

$$I(\hat{P}_{(x^n, y^n)}) = D(\hat{P}_{(x^n, y^n)} \| [\hat{P}_{x^n} \cdot \hat{P}_{y^n}]) = \sum_{a \in \mathcal{X}} \sum_{b \in \mathcal{Y}} \hat{P}_{(x^n, y^n)}(a, b) \log \frac{\hat{P}_{(x^n, y^n)}(a, b)}{\hat{P}_{x^n}(a) \hat{P}_{y^n}(b)}$$

where $\hat{P}_{x^n}$ and $\hat{P}_{y^n}$ are empirical distributions of x^n and y^n , respectively. The idea of the MMI decoder is that if X^n and Y^n are generated independently, then their empirical distribution $\hat{P}_{(x^n, y^n)}$ should be close to the product of two marginals $\hat{P}_{x^n}(a)$ and $\hat{P}_{y^n}(b)$ and hence the empirical mutual information should be close to 0.

Now consider a communication setup as follows. Generate i.i.d. random codebook with e^{nR} codewords $\{X^n(1), \dots, X^n(e^{nR})\}$ from a distribution P_X . Transmit the first the codeword $x^n(1)$ by n -uses of a DMC(W). Upon receiving $Y^n = y^n$, the MMI decoder computes for each codeword $x^n(m)$, $m = 1, 2, \dots, e^{nR}$, the empirical mutual information of $(x^n(m), y^n)$ and denote the corresponding value by $S(m)$, i.e.,

$$S(m) \triangleq I(\hat{P}_{(x^n(m), y^n)}).$$

If there exists a unique m such that $S(m) \geq I(P_X, W) - \epsilon > 0$ for a fixed $\epsilon > 0$, the decoder claims that $\hat{M} = m$, and otherwise the decoder claims an error. The error probability is then

$$P_e^{(n)} = \Pr(\hat{M} \neq M) \leq \Pr(\mathcal{E}_1) + \Pr(\mathcal{E}_2)$$

where

$$\mathcal{E}_1 \triangleq \{\text{For the correct message } m = 1, S(1) < I(P_X, W) - \epsilon\}$$

$$\mathcal{E}_2 \triangleq \{\exists m \neq 1 \text{ such that } S(m) \geq I(P_X, W) - \epsilon > 0\}$$

- a) (5 points) Show that $\Pr(\mathcal{E}_1) \rightarrow 0$ as $n \rightarrow \infty$. (Assume that $P_{XY}(a, b) = P_X(a) \cdot W(b|a) > 0$, $\forall a \in \mathcal{X}$, $\forall b \in \mathcal{Y}$, and $|\mathcal{X}|$ and $|\mathcal{Y}|$ are finite.)
 Hint: You may use the fact that $P_{XY}^n(\tilde{T}_{P_{XY}}) \rightarrow 1$ for any $\delta > 0$ where

$$\tilde{T}_{P_{XY}} = \{(x^n, y^n) : |\hat{P}_{(x^n, y^n)}(a, b) - P_{XY}(a, b)| \leq \delta, \forall a \in \mathcal{X}, \forall b \in \mathcal{Y}\},$$

and for $(x^n, y^n) \in \tilde{T}_{P_{XY}}$ it is guaranteed that $x^n \in \tilde{T}_{P_X}$ and $y^n \in \tilde{T}_{P_Y}$ where

$$\begin{aligned}\tilde{T}_{P_X} &= \{x^n : |\hat{P}_{x^n}(a) - P_X(a)| \leq \delta|\mathcal{Y}|, \forall a \in \mathcal{X}\}, \\ \tilde{T}_{P_Y} &= \{y^n : |\hat{P}_{y^n}(b) - P_Y(b)| \leq \delta|\mathcal{X}|, \forall b \in \mathcal{Y}\}.\end{aligned}$$

Answer:

b) (10 points) Show that $\Pr(\mathcal{E}_2) \rightarrow 0$ as $n \rightarrow \infty$ when $R < I(P_X, W) - \epsilon$.

Hint: You may use Sanov's theorem and the following fact. Consider the following set of joint distributions

$$\mathcal{A}_\gamma \triangleq \{Q_{XY} : I(Q_{XY}) \geq \gamma\}.$$

For any Q_{XY} , we have

$$\begin{aligned} D(Q_{XY} \| [P_X \cdot P_Y]) &= \sum_{a \in \mathcal{X}} \sum_{b \in \mathcal{Y}} Q_{XY}(a, b) \log \frac{Q_{XY}(a, b)}{P_X(a)P_Y(b)} \\ &= \sum_{a \in \mathcal{X}} \sum_{b \in \mathcal{Y}} Q_{XY}(a, b) \left[\log \frac{Q_{XY}(a, b)}{Q_X(a)Q_Y(b)} + \log \frac{Q_X(a)Q_Y(b)}{P_X(a)P_Y(b)} \right] \\ &= I(Q_{XY}) + D(Q_X \| P_X) + D(Q_Y \| P_Y) \\ &\geq I(Q_{XY}) \end{aligned}$$

Thus,

$$\min_{Q_{XY} \in \mathcal{A}_\gamma} D(Q_{XY} \| [P_X \cdot P_Y]) \geq \gamma.$$

Answer:

Problem 3

In class, we derived the random coding error exponent $E_r(R)$ for a BSC(ϵ) for $\epsilon < 1/2$. We showed that the average decoding error probability $P_e^{(n)} \stackrel{\triangle}{\leq} e^{-nE_r(R)}$ for $R < C$ and derived $E_r(R)$ for every $R < C$. In particular, for low rate regimes $R < R_{\text{crit}}$, we showed that $E_r(R) = E_0 - R$ where

$$E_0 = \min_{\gamma} D_{\text{B}}(\gamma||\epsilon) + D(\gamma||1/2) = \log 2 - \log(1 + 2\sqrt{\epsilon(1-\epsilon)}). \quad (1)$$

In the high rate regime $R > R_{\text{crit}}$, we showed that

$$E_r(R) = D_{\text{B}}(\gamma_{\text{GV}}(R)||\epsilon) \quad (2)$$

where $\gamma_{\text{GV}}(R)$ satisfies $R = D_{\text{B}}(\gamma_{\text{GV}}(R)||1/2)$. In this problem, we will compare E_0 (the random coding error exponent at rate $R = 0$) with the error exponent for two repetition codes $\underbrace{00 \dots 0}_n$ and $\underbrace{11 \dots 1}_n$.

- a) (5 points) Assume that one of two repetition codes $\underbrace{00 \dots 0}_n$ and $\underbrace{11 \dots 1}_n$ is transmitted by n uses of a BSC(ϵ) for $\epsilon < 1/2$. The ML (maximum likelihood) decoder claims that 0 was transmitted if the received number of 0's are larger than $n/2$ and it claims 1 otherwise. Calculate the error exponent E_{rep} of the corresponding decoding error probability $P_e^{(n)} \doteq e^{-nE_{\text{rep}}}$ as $n \rightarrow \infty$.

Answer: $E_{\text{rep}} =$

Reasoning for Problem 3(a):

- b) (5 points) Compare the error exponent E_{rep} for repetition coding (your answer in a) with E_0 in (1) (the error exponent of random coding at rate $R = 0$). Which one is bigger? (Circle your answer)

Answer: $E_{\text{rep}} \geq E_0$ $E_{\text{rep}} \leq E_0$

Reasoning for Problem 3(b): Provide the mathematic proof for your answer.

- c) (5 points) Provide intuitive explanation on your answer in b). If your answer is $E_{\text{rep}} \geq E_0$, explain why repetition coding is better than random coding when we transmit one of two codewords. If your answer is $E_{\text{rep}} \leq E_0$, explain why random coding is better than repetition coding when we transmit one of two codewords.

Reasoning for Problem 3(c):

See the next page for the Bonus problem!

- d) (Bonus problem: 5 points) In class, we found a lower bound $E_r(R)$ on the error exponent $E^*(R)$ by showing that $P_e^{(n)} \stackrel{\dot{\leq}}{e^{-nE_r(R)}}$ when we assume that encoder uses random coding and decoder uses the ML decoding rule. In this problem, let's discuss an idea of deriving an upper bound $E_s(R)$ on the error exponent $E^*(R)$ by showing that with any encoder and decoder $P_e^{(n)} \stackrel{\dot{\geq}}{e^{-nE_s(R)}}$ at a given rate R . By deriving the upper bound, we can show that $E_r(R) \leq E^*(R) \leq E_s(R)$.

Suppose that we communicate at a rate R lower than the capacity of the BSC(ϵ), i.e., $R < C = \log 2 - H_B(\epsilon)$. The channel output $Y^n = X^n \oplus Z^n$ where the channel noise sequence Z^n is i.i.d. Bern(ϵ). Consider the situation where the channel behaves atypically and it flips $n\gamma$ bits of the transmitted codewords for $\gamma > \epsilon$, i.e., the empirical distribution of Z^n is Bern(γ). If the communication rate R is larger than the capacity of the realized channel BSC(γ), i.e., $R \geq \log 2 - H_B(\gamma)$ which is equivalent to $\gamma \geq \gamma_{GV}(R)$, it is clear that the decoding error cannot go to 0 by the converse of the channel coding theorem. This implies that

$$P_e^{(n)} \stackrel{\dot{\geq}}{\Pr(|Z^n| \geq n\gamma_{GV}(R))}$$

where $|Z^n| = \sum_{i=1}^n \mathbb{1}(Z_i = 1)$. Calculate $\Pr(|Z^n| \geq n\gamma_{GV}(R)) \doteq e^{-nE_s(R)}$ up to exponential approximation and find $E_s(R)$. Compare this upper bound $E_s(R)$ with the lower bound $E_r(R)$ in (2) for $R > R_{crit}$ and provide discussion if you found any interesting points.

Answer: $E_s(R) =$

Reasoning for Problem 3(d):